

SURYA SINGU

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EXPERIENCE

Developer Content Creator & Community Builder | Instagram: @surya_code

Sep 2025 – Present

Remote

- Grew a **18.5K-follower developer community** in under 6 months on Instagram covering blockchain, Web3, and AI — posts reaching **1M, 420K, and 343K** organic views, proving rapid ability to build and engage a technical audience at scale.
- Produce **educational tech content** (39+ posts) simplifying complex developer topics — directly mirroring the DevRel mandate of driving ecosystem adoption through accessible education and **grant proposal** awareness.
- Demonstrated ability to **build, grow, and engage** a developer audience from scratch — the core skill needed to expand builder communities across ecosystems and beyond.

Palo Alto Networks

Apr 2025 – Jun 2025

Cybersecurity Intern (Virtual)

Remote

- Completed structured training across Cybersecurity Fundamentals, Cloud Security, Network Security, and SOC operations; applied threat detection and incident response procedures to real-world scenarios.

PROJECTS

Foundry Smart Contract Lottery | Solidity, Foundry, Chainlink VRF v2.5, Ethereum | GitHub

Feb 2026

- Architected a fully decentralized, trustless lottery on Ethereum eliminating any central authority — winner selection is governed entirely by **Chainlink VRF v2.5**, making outcomes cryptographically provable and manipulation-proof.
- Automated the entire draw lifecycle using **Chainlink Automation** (`checkUpkeep/performUpkeep`), removing the need for any manual trigger and ensuring time-based draws execute autonomously on-chain.
- Prevented reentrancy and double-spend exploits by rigorously applying the **CEI (Checks-Effects-Interactions)** pattern and strict access control across all state-changing functions.
- Achieved high test reliability by writing **unit and integration tests** (`RaffleTest.t.sol`, `Interactions.t.sol`) with Foundry, tracking coverage metrics in `coverage.txt` to identify and close gaps.
- Reduced gas costs using **Solmate** gas-optimized contract libraries; managed full dev lifecycle via **Makefile** automation for build, test, and Sepolia testnet deployment.
- Modularized scripts into `DeployRaffle.s.sol`, `HelperConfig.s.sol`, and `Interactions.s.sol` for clean separation of deployment, network config, and subscription management.

Zero Knowledge Proof Cryptographic System | Python, SHA-256, Schnorr Protocol, Streamlit | GitHub

Jan 2026

- Designed and built a production-quality **Schnorr-like ZKP authentication system** from scratch using 256-bit secp256k1 modular arithmetic and SHA-256 — proving password knowledge without the secret ever leaving the client.
- Achieved **sub-millisecond cryptographic performance**: < 1ms proof generation and < 1ms verification per round, validated across 256-bit prime modular exponentiation and logged to `results.csv`.
- Extended the system to a **digital forensics use case** — verifying file authenticity for court evidence and chain-of-custody without exposing file contents, demonstrating real-world security applicability.
- Formally satisfied all three ZKP security properties: **Completeness** (honest prover always succeeds), **Soundness** (forging without the secret has probability $\approx 1/p \approx 0$), and **Zero Knowledge** (verifier learns nothing about the secret).
- Architected a **6-layer modular system** separating prover, verifier, authentication, forensics, performance, and UI — enabling independent testing and future extension of each cryptographic component.
- Shipped a **Streamlit web UI** and a CLI runner with live attack simulation demos contrasting traditional credential transmission against the ZKP approach, making the security advantage immediately visible.

TECHNICAL SKILLS

Blockchain & Web3: Solidity · Ethereum · EVM · Web3.js · DApp Development · ERC-20/721/4337 · Token Standards · Consensus Mechanisms · Hyperledger

Languages: Solidity · TypeScript · JavaScript · Python · Java · C

Tools & Frameworks: Foundry (Proficient) · Chainlink VRF · Chainlink Automation · Node.js · Git / GitHub · Hardhat (Familiar)

Cryptography & Security: ZKPs · SHA-256 · secp256k1 · Keccak256 · Modular Arithmetic · Network Security · IDS

Dev Relations: Technical Writing · Documentation · Developer Education · Community Building · Workshop Facilitation

EDUCATION

GITAM Deemed University

Graduating April 2026

B.Tech., Computer Science Engineering — Bengaluru, IN

CGPA: 8.08 / 10.0